

SNAPSHOT AI

Private On-Device Scam & Phishing Detector

A compact AI product concept designed around privacy, explainability and Snapdragon-powered PC deployment.

PRIVATE Local-first message analysis	EXPLAINABLE Reasons beside the risk score
LIGHTWEIGHT Small local inference pipeline	SNAPDRAGON-READY ONNX / Qualcomm execution path

1. THE PROBLEM

Scam and phishing messages use urgency, impersonation, payment requests and suspicious links. Users often have to decide quickly whether a message is safe, while the message itself may contain private information.

2. THE SOLUTION

SnapShield AI accepts a message, email snippet, chat text or URL and analyzes it locally. A lightweight classifier looks for language patterns while a transparent security layer checks URL and sensitive-information signals. The result is LOW, MEDIUM or HIGH risk with reasons.

3. PRODUCT EXPERIENCE

The interface is intentionally simple: paste → analyze → understand. The user sees the result and the signals behind it instead of receiving an unexplained number.

4. TECHNICAL DESIGN

Python desktop UI → local TF-IDF text features → Logistic Regression → URL/security heuristics → explainable risk result.

The inference boundary is kept compact so the model can be represented in ONNX and connected to the Windows-on-Snapdragon execution stack. This makes the concept suitable for Snapdragon-powered HP PC deployment.

5. DIFFERENTIATION

The project combines privacy-first local processing with explainable warnings and a clear hardware-aware deployment path. The same inference core can later support browser, email and messaging workflows.

6. IMPACT

The intended value is not just a classification label. It is a small intervention that gives a user a reason to pause before sharing credentials, paying a fee or opening a suspicious link.

7. IMPLEMENTATION

The repository contains a working desktop prototype, demonstration dataset, architecture notes, and a short demo script. The prototype uses no cloud API.

8. SNAPDRAGON DEPLOYMENT PATH

The deployment path is: local model → ONNX representation → ONNX Runtime / Qualcomm execution → Snapdragon-powered HP PC. The target product can then be profiled for latency, memory and energy use.

9. VALIDATION & RESPONSIBLE CLAIMS

The included data is demonstration data, not a production security corpus. Production development would require a properly licensed representative dataset, adversarial testing, security review and device-level profiling. No unmeasured hardware performance is presented as a result.

10. DELIVERABLES

Source code • README • architecture • demo script • designed proposal PDF • 12-slide pitch PPTX • matching pitch PDF